Activation-Specific Signatures of Tian's General Repulsion Mechanism

(Technical summary)

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1. Problem

Tian [2025] proves a repulsion theorem (Theorem 6) for two-layer networks $\widehat{Y} = \sigma(XW)V$ ($\sigma(x)$=x^2, MSE loss with sample-mean projector $P_1^{\perp}$) during the interactive feature-learning stage of grokking. The asserted off-diagonal sign rule for $B := (\widetilde{F}^\top \widetilde{F} + \eta I)^{-1}$:

$$\operatorname{sgn}(B_{j\ell}) \;=\; -\operatorname{sgn}\!\Big(\widetilde{f}_j^\top P_{\eta,-j\ell}\widetilde{f}_\ell\Big), \qquad P_{\eta,-j\ell} := I - \widetilde{F}_{-j\ell}(\widetilde{F}_{-j\ell}^\top \widetilde{F}_{-j\ell} + \eta I)^{-1}\widetilde{F}_{-j\ell}^\top. \tag{1}$$

Two questions the framework does not answer empirically: (i) when does this become observable in training, and (ii) does it generalize beyond the $\sigma(x)$=x^2 regime in which it is proven?

2. Result

The mechanism is activation-general; its spectral signature is activation-specific.

Theorem 6 verification across activations. Computing B via the Woodbury identity,

$$B = \tfrac{1}{\eta} I - \tfrac{1}{\eta^2}\widetilde{F}^\top(\widetilde{F}\widetilde{F}^\top + \eta I)^{-1}\widetilde{F} \quad (K \times K \text{ inverse} \to n \times n, \text{ float64 on CPU}), \tag{2}$$

on top-200 most-similar feature pairs at five deterministic-replay checkpoints, multi-seed sign-match progression on Tian's exact setup ($M = 71$, $K = 2048$, $n = 2016$, $\eta = 2{\times}10^{-4}$):

epoch	50	100	**175 (lock-in)**	250	300	range
$\sigma = x^2$ ($n = 5$ seeds)	0.865	0.895	**0.955**	0.970	**0.985**	IQR ≤ 0.015
σ = ReLU (seed 0)	—	0.91	—	—	0.995 (ep 300), 1.000 (ep 500)	—

The $P_{\eta,-j\ell} \approx P_\eta$ approximation preserves the sign rule on directly-checked pairs (10/10 sign agreement at epoch 175 seed 0); the magnitude of the residual differs because the full projector P_η nearly annihilates $\widetilde{f}_\ell$ while $P_{\eta,-j\ell}$ does not, but for the sign rule of equation (1) the approximation is sufficient.

Parameter-update spectral signature. At each epoch we form the rolling-window Gram of ΔW ($W = 20$, $P = 2MK$):

$$\Delta := [\Delta W_{t-W+1}, \ldots, \Delta W_t] \in \mathbb{R}^{P\times W}, \quad \sigma_k(t) := \lambda_k(\Delta^\top \Delta). \tag{3}$$

A slope-based detector,

$$s(t) := \tfrac{1}{25}\big[\log(\sigma_2/\sigma_3)(t) - \log(\sigma_2/\sigma_3)(t-25)\big], \quad \text{"fire"} := \min\{t \geq 100 : s(t) > 0.04\}, \tag{4}$$

gives, on $\sigma = x^2$ at the headline operating point ($\eta = 2{\times}10^{-4}$ vs $\eta = 0$, 15 seeds each):

- **15/15 grok seeds fire at epoch 174 (IQR $[173, 174]$); 0/15 control seeds.** Late-stage magnitude separation 229× ($\sigma_2/\sigma_3 \approx 300$ grok vs ≈ 1.31 control over epochs 200–400). The slope-fire epoch coincides with Theorem 6 saturation (≥ 0.95).
- Mechanism: between epochs 150 and 200, σ_2 stabilizes near 5×10^{-3} while σ_3 collapses two orders of magnitude. The rolling spectrum becomes effectively rank-2.

On $\sigma = \text{ReLU}$ (15 seeds, 800 epochs, otherwise identical):

- **0/15 grok seeds fire.** Late-stage magnitude separation collapses to 1.4×. The spectrum is rank-1 dominated: $\sigma_1 \gg \sigma_2 \approx \sigma_3 \approx \sigma_4 \approx \sigma_5$.

3. Interpretation: structure vs mechanism

Tian's Theorem 6 governs the off-diagonal sign structure of B via properties of $\widetilde{F}^\top \widetilde{F}$ alone, which depends on the activation only through its effect on $\widetilde{F}$. The rolling-ΔW spectral signature of Theorem 6's saturation depends on *how* the repulsion translates into weight updates, which depends on σ':

- Power activations ($\sigma(x)=x^2$) produce *focused memorization* (Tian, 2025 Theorem 5): features collapse onto sharp peaks. Theorem 6 repulsion consolidates redundant features onto two persistent rank-2 directions in ΔW; we observe this directly.
- ReLU/sigmoid activations produce *spreading memorization*: features remain distributed. Theorem 6 still operates (sign rule still holds), but there are no sharp peaks for repulsion to consolidate around; ΔW remains rank-1 dominated.

The two findings together identify which parts of the `Li`$_2$ framework are activation-general (Theorem 6's sign rule itself) versus activation-specific (the rolling-ΔW rank-2 signature).

4. Methodological controls

Window-size sensitivity ($n = 3$ seeds, $\sigma = x^2$):

W	grok fire (3 seeds)	ctrl fire (3 seeds)	late σ_2/σ_3 grok	late σ_2/σ_3 ctrl
5	[144, 149, 153]	**[179, 193, 263] (3/3 fire)**	408	8.0
10	[—, —, 347] (1/3)	**[263, 266, 278] (3/3)**	64	2.3
20	[173, 173, 174]	[—, —, —] (0/3)	285	1.3
30	[180, 180, 180]	[—, —, —] (0/3)	140	1.2

$W \in \{20, 30\}$ give perfect specificity; $W \leq 10$ produce false positives in control. The transition is ~ 25 epochs wide, so the window must average over this without smearing into Stage I/II.

Rank confirmation via σ_4, σ_5: At $W \leq 10$, $\sigma_3, \sigma_4, \sigma_5$ collapse together to $\sim 10^{-5}$ (exact rank-2). At $W = 30$, geometric cascade $\sigma_3 \approx 10^{-4}$, $\sigma_4 \approx 10^{-5}$, $\sigma_5 \approx 10^{-6}$ (rank $\lesssim 3$). The "rank-2" framing is exact at the finest temporal resolution.

5. Extension across η and the level-metric initiation detector

η sweep ($n = 25$ runs, $\sigma = x^2$). An extended single-seed run at $\eta = 10^{-5}$ to 2000 epochs confirms Tian's $1/\eta$ scaling: grokking at epoch 1527, with the level metric

$$\rho_{\text{tian}}(t) := \frac{\left\| P_1^\perp F F^\top - (a(t) I + b(t)\, \mathbf{1}\mathbf{1}^\top) \right\|_F}{\|FF^\top\|_F} \tag{5}$$

crossing 0.075 at epoch 527 (lead 567 epochs vs $t_{\text{test}=0.5}$). At larger η the lead is positive in 20/20 grokking runs (range 79–154 epochs).

The lock-in magnitude is η-dependent: at $\eta = 10^{-5}$ peak $\sigma_2/\sigma_3 \approx 25$ vs ≈ 300 at $\eta = 2\times10^{-4}$. The rank-2 structure is incompletely developed when measurement ends in the slow regime, but its existence is preserved.

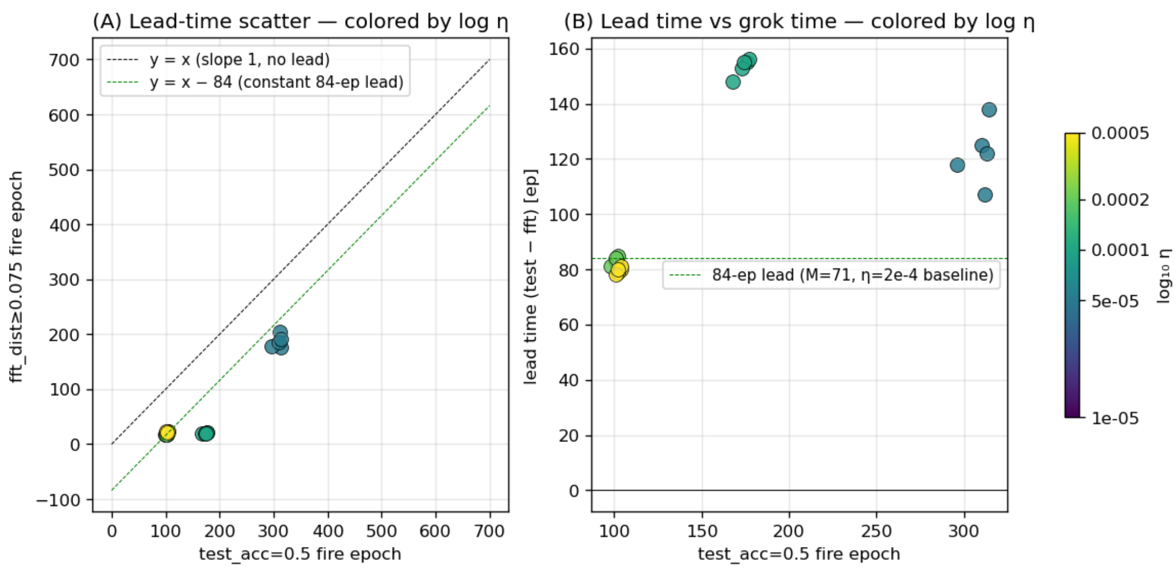
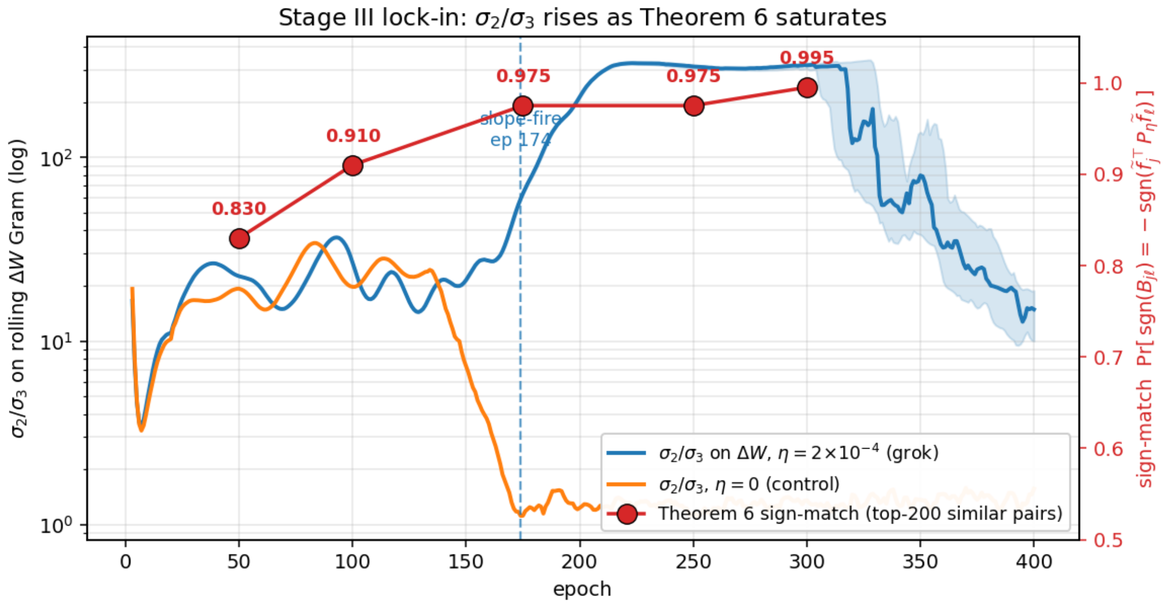


Figure 1: η sweep ($n = 25$, $\sigma = x^2$): ρ_{tian} fire epoch versus $t_{\text{test}=0.5}$. Lead positive in 20/20 grokking runs; lead time scales with grokking timescale. Linear regression $t_\rho = 0.80\, t_{\text{test}} - 75$ ($R^2 = 0.87$).

Tightly-scoped applicability. ρ_{tian} is a Stage I→II initiation marker but does NOT survive an $M \times p$ scaling sweep: on a 60-run sweep ($M \in \{41, 71, 127\}$, $p \in \{0.1, 0.2, 0.3, 0.5\}$, 5 seeds), it fires in **60/60 runs** regardless of grokking outcome. Nor does it survive activation change: on $\sigma = \text{ReLU}$ it fires at epoch 0 because the ReLU initialization is already far from the form $aI + b\mathbf{1}\mathbf{1}^\top$ (asymmetric activation ⇒ non-isotropic random-feature Gram). The detector is scoped to activations producing approximately isotropic random features at init.

6. Figure



Blue: median σ_2/σ_3 on rolling ΔW Gram across 15 grok seeds at $\sigma = x^2$, $\eta = 2\times10^{-4}$, with IQR shading and the slope-fire epoch (174) marked. **Orange:** median across 15 control seeds. **Red:** Theorem 6 sign-match on top-200 most-similar feature pairs at five deterministic-replay checkpoints (median across $n = 5$ seeds, IQR ≤ 0.015 throughout). The σ_2/σ_3 rise to its post-grokking plateau coincides exactly with Theorem 6 sign-match crossing 0.95. The same sign rule holds on $\sigma = \mathrm{ReLU}$ (sign-match $\to$ 1.000 by epoch 500), but the rolling-ΔW rank-2 signature does not (separation collapses from 229× to 1.4×).

References

Yuandong Tian. Provable scaling laws of feature emergence from learning dynamics of grokking. *arXiv preprint arXiv:2509.21519*, 2025. URL `https://arxiv.org/abs/2509.21519`.